H–B–292a
June 3, 1963

SUPERSEDING
Emer. Fed. Spec. H-B-0292
August 22, 1951

FEDERAL SPECIFICATION

BRUSH, ADHESIVE, (GLUE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers flat and round glue brushes of natural hog bristle.

1.2 Classification.

1.2.1 *Types and sizes.* Glue brushes shall be of the following types and sizes, as specified (see 6.2):

Type F.—Flat (see figure 1).
Size.—1 inch.
Size.—2 inch.
Size.—3 inch.

Type R.—Round (see figure 1.).
Size.—13/16 inch.
Size.—1 inch.
Size.—1-3/16 inch.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

FSC 7920

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Standard product.** The brushes delivered under this specification shall be the manufacturer's standard commercial product, except for any design modifications and additional requirements specified herein.

3.2 **Preproduction sample.** When specified (see 6.2), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

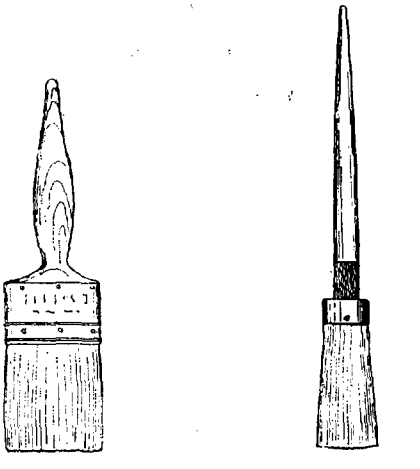
3.3 **Materials.** Materials shall conform to the applicable specifications and requirements specified hereinafter. Materials not specifically defined shall be of the quality normally used for the purpose in the manufacturer's standard commercial practice.

3.3.1 *Wood.* The wood shall be close-grained and straight-grained hardwood such as beech, birch, or maple.

3.3.2 *Setting compound.* Bristle setting compound shall be vulcanized hard rubber, epoxy resin, or other suitable setting material, which shall not become brittle or be affected by materials with which the brushes are normally used.

3.3.3 *Bristle.* The bristle shall be straight, resilient, new, natural hog bristle with no artificial coloring matter added and shall be cleaned and sterilized as specified in 3.5. When tested as specified in 4.4.3, there shall be no perceptible change in stiffness. When tested as specified in 4.4.4, there shall be no evidence of artificial color.

3.4 **Construction.** The brushes shall be similar to figure 1 and shall conform to the requirements specified herein and in tables II and III as applicable. The bristle shall conform to the requirements specified in 3.3.3 and table I. The butt end of all bristle shall be set in setting compound as specified in 3.3.2. When tested as specified in 4.4.1, it shall not be possible to remove the bristles from the brush part of the setting compound without breaking the bristles. The bristle weight shall be as shown in tables II and III as applicable, when tested as specified in 4.4.2.2. The brushes shall be trimmed to a square top.



TYPE F FLAT TYPE R ROUND

FIGURE 1.—Glue brushes.

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TABLE I - Bristle

Type F, size	Type R, Size	Percent top size (full length)
1 inch	13/16 inch	60
2 inch	1 inch	70
3 inch	1-3/16 inch	70

TABLE II.- Physical requirements for type F brushes

		Size 1-inch	Size 2-inch	Size 3-inch
Brush part:				
Width inside ferrule	inches	1 ± 1/16	2 ± 1/16	3 ± 1/16
Thickness inside ferrule	inch	7/16 ± 1/16	9/16 ± 1/16	11/16 ± 1/16
Bristle				
Weight in finished brush, minimum	ounces	7/16	1-1/4	2-1/4
Hold in ferrule, minimum	inch	3/8	3/8	3/8
Length outside of ferrule, minimum	inches	2-1/8	2-5/8	2-7/8
Ferrule:				
Depth, minimum	inches	1-1/4	1-1/4	1-1/4
Thickness, minimum	inch	0.012	0.012	0.012

TABLE III.- Physical requirements for type R brushes

		Size 13/16-inch	Size 1-inch	Size 1-3/16 inch
Brush part:				
Diameter inside ferrule	inches	13/16 ± 1/32	1 ± 1/32	1-3/16 ± 1/32
Bristle				
Weight in finished brush, minimum	ounces	3/8	5/8	1-3/32
Hold in ferrule, minimum	inch	1/2	1/2	1/2
Length outside of ferrule, minimum	inches	2	2-1/2	2-3/4
Ferrule: Thickness, minimum	inch	0.042	0.042	0.042

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3.4.1 *Type F.* The type F glue brushes shall have a panel or beaver tail type handle constructed of wood specified in 3.3.1 and finished as specified in 3.6. The brushes shall have a steel ferrule made corrosion resisting by a suitable coating which shall not be affected by materials with which the brushes are normally used. The ferrule shall have an end lap of not less than 1/4 inch, securely soldered and reinforced by not less than two rivets, or shall be interlocked, lockseamed, or welded. The ferrule shall be riveted or through-nailed and clinched, using one rivet or nail in size 1-inch brushes, two rivets or nails in size 2-inch brushes, and three rivets or nails in size 3-inch brushes. One nail or rivet shall penetrate both thicknesses of the lap if the lap is on the flat side. If the lap is on the end, the rivets or nails shall be spaced 1 inch apart. When epoxy resin is used as a setting compound, riveting or nailing shall not be required.

3.4.2 *Type R.* The type R glue brushes shall have a plain, round wooden handle, constructed of wood specified in 3.3.1, with a smooth sanded finish. The brushes shall have a seamless steel ferrule made corrosion resisting by a suitable coating which shall not be affected by materials with which the brushes are normally used.

3.5 **Sterilization.** Hog bristles shall be sterilized prior to the manufacturing process, after thorough washing and rinsing, in accordance with either 3.5.1, 3.5.2, or 3.5.3. A properly checked steam-pressure gauge (applicable for the method in 3.5.3) and a thermometer (for all methods), both visible from the outside of the chambers, shall be provided with a suitable recording device to provide a record of the pressure, when applicable, and temperature of the disinfecting chamber.

3.5.1 *Boiling water.* Hog bristle shall be soaked in boiling water (212°F.) for not less than 30 minutes.

3.5.2 *Dry heat.* Hog bristles shall be placed in a chamber and a temperature above 284°F. shall be developed and maintained for not less than 3 hours.

3.5.3 *Autoclaving.* Hog bristles shall be placed in a steam-pressure chamber for not less than 45 minutes at not less than 15 p.s.i.g.

3.6 **Finish.** The type F handles shall be coated with varnish, lacquer, or other suitable clear organic material in accordance with the manufacturer's standard commercial practice.

3.7 **Marking for identification.** In addition to any special marking required by the contract or order (see 6.2), the contractor shall permanently indent his name, or trademark of such known character easily identifiable with said manufacturer, on the handle or ferrule of each brush. Marking located on the handle shall be coated with a commercial, waterproof stencil paint.

3.8 **Workmanship.** The finished brushes shall be clean, well made, and shall be free from any defects which may affect appearance or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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TABLE IV.- Instructions for intermediate testing

Characteristic	Specification reference		Number of determinations per unit
	Requirement	Test method	
Weight of bristle	3.4 and tables II & III	4.4.2	1
Stiffness of bristle	3.3.3	4.4.3	10
Color of bristle	3.3.3	4.4.4	1

4.2 **Preproduction sample inspection.** The preproduction sample shall be examined for defects specified in 4.3.4.1.1, dimensions specified, and shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3 and 4.4.4.

4.3 **Sampling and inspection for acceptance.** Sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the same type and size offered for delivery at one time.

4.3.1 *Inspection of components and materials.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.2 *Process examination.* Examination shall be made of the sterilization of the hog bristle to establish that no deviation is made from the specified procedure or requirement. Whenever a deviation is noted, correction shall be made.

4.3.3 *Intermediate testing.* When epoxy resin is used as a setting compound, intermediate tests shall be performed for the applicable characteristics indicated in table IV herein, prior to the application of the setting compound. The sample unit shall be the amount of bristle to be used in one brush and the lot size shall be expressed in units of brushes. The requirements shall be applicable to the individual unit and the results shall be reported as pass or fail. The inspection level shall be level L3 and acceptable quality level shall be 4.0.

4.3.4 *Inspection of the end item.*

4.3.4.1 *Examination of the end item.* Examination of the end item shall be in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of brushes for examination under 4.3.4.1.1 and 4.3.4.1.2, in units of intermediate containers for examination under 4.3.5.1 and in units of shipping containers for examination under 4.3.5.2.

4.3.4.1.1 *Examination of the end item for defects in material, construction and workmanship.* The sample unit for this examination shall be one brush.

Examine	Defect
Construction and appearance, type F	Handle not panel or beaver tail shaped.
	Wood component not finished with varnish, lacquer, or suitable clear organic material.

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Examine	Defect
Construction and appearance, type F (cont'd)	Ferrule, end lap less than 1/4 inch, not securely soldered, not reinforced by two or more rivets; not interlocked, not lockseamed, or not welded, (as applicable).
	Not riveted, or not through-nailed and clinched, (as applicable).
	No rivet or nail in size 1-inch brush (as applicable).
	Less than 2 rivets or nails in size 2-inch brush (as applicable).
	Less than 3 rivets or nails in size 3-inch brush (as applicable.)
	One rivet or nail not penetrating both thickness of lap for flat lap, or rivets or nails not spaced 1 inch apart for end lap, (as applicable).
Type R	Handle not plain and round.
	Ferrule not seamless.
Workmanship	Brush filler material not evenly distributed in brush head.
	Brush filler material not clean; evidence of dirt or foreign matter.
	Brush head not evenly trimmed.
	Wood component not smoothly finished; splintered, torn, or loose grain.
Marking	Identification marking omitted, incomplete, incorrect, or not as specified.
	Not coated with waterproof stencil paint.

4.3.4.1.2 *Dimensional examination.* Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension not within the specified requirements shall be classified as a defect. The sample unit for this examination shall be one brush.

4.3.4.1.3 *Inspection levels and acceptable quality levels (AQL's) for examinations.* The inspection levels for determining the sample size and the acceptable quality levels (AQL's), expressed in defects per hundred units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.4.1.1	I	4.0
4.3.4.1.2	L2	2.5
4.3.5.1	L4	Acceptance No.—0 Reject No.—1
4.3.5.2	L4	4.0

4.3.4.1.4 *Testing of the end item.* The end item shall be tested for the applicable characteristics as indicated in table V herein from each lot presented for examination of each type and size brush. The sample unit shall be one brush and the lot size shall be expressed in units of brushes. The requirements shall be applicable to the individual unit and the results reported as pass or fail. The inspection level shall be L3 and acceptable quality level shall be 4.0.

4.3.5 *Examination of preparation for delivery.*

4.3.5.1 *Examination of the intermediate container for quantity.* The sample unit for this examination shall be one intermediate container. Not less than three sample units shall be selected for this examination. The quantity per container shall be not less than specified.

4.3.5.2 *Examination of preparation for delivery (except count).* An examination shall be made to determine that packaging, packing and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified below. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging	Not unit packaged as specified.
Packing	Any packing component or material missing, damaged, or not as specified; closure not accomplished by specified or required methods or materials.
Workmanship	Inadequate application of component or material such as incomplete closure of container flap, tape, or caseliner.
	Bulge or distortion of container.
Content	Number of unit or intermediate packages not in accordance with requirements.
Marking	Omitted, incomplete, incorrect, illegible; not in accordance with contract requirements.

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TABLE V.- Instructions for testing

Character	Specification Reference		Number Determinations Per Unit
	Requirement	Test method	
Tightness of bristle	3.4	4.4.1	10
Weight of bristle *	3.4 and tables II & III	4.4.2	1
Stiffness of bristle *	3.3.3	4.4.3	10
Color of bristle *	3.3.3	4.4.4	1

* This test to be performed on the end item when 4.3.3 does not apply.

4.4 Tests.

4.4.1 *Bristle tightness.* Individual strands of bristle shall be held between the thumb and forefinger and pulled to determine compliance with 3.4.

4.4.2 *Bristle weight.*

4.4.2.1 *Removal of bristle.* When epoxy resin is not used as a setting compound, the bristle shall be removed from the brush. The bristle shall be removed by immersing the setting compound and the butt end of the bristles in gasoline or other suitable solvent to a depth of not more than 1/8 inch above the setting compound. After the setting compound has been removed, the solvent shall be allowed to evaporate and, where applicable, the thread shall be removed from the knotted tufts.

4.4.2.2 *Conditioning of bristle.* The bristle removed from the brush as specified in 4.4.2.1 or as specified in 4.3.3 shall be conditioned for not less than 16 hours in air at a relative humidity of 50 plus or minus 4 percent and at a temperature of 23.0° ± 1.1°C. Weighing to determine compliance with 3.4 shall be performed in the same atmosphere.

4.4.3 *Stiffness test.* The bristle tested as specified in 4.4.2.2 shall be soaked in boiling water (212°F.) for not less than one hour. The bristle shall be removed from the water and placed face up in a holding device so that it is held firmly in a manner and position similar to a finished brush. A finished brush shall be attached in a face up position alongside the bristle. The bristle in the holding device and the brush shall be depressed with the palm of the hand not less than 10 times respectively for comparison to determine compliance with 3.3.3.

4.4.4 *Color test.* The bristle tested as specified in 4.4.3 and the water used for the test shall be examined for compliance with 3.3.3.

5. PREPARATION FOR DELIVERY

(For civil agency procurements, Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit package.* Each brush shall be individually wrapped in a double thickness of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. The ends of the paper wrapper shall be folded back over each end of the brush. The wrapper shall then be secured with tape conforming to class 1 of UU-T-111.

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5.1.1.2 *Intermediate package.* Twelve brushes of one type and size only, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting folding or set-up paperboard box conforming to style III, type B, class a, or style IV, type A or B, class a of PPP-B-566; or type I or II, grade A, class A, style 2 of PPP-B-676, respectively. The brushes shall be arranged in an alternating bristle-to-handle order. Not less than 3 grams of naphthalene in ball or flake form shall be placed in each box. Each box shall be closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Brushes shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* One hundred and forty-four brushes of one type and size only, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 *Level B.* One hundred and forty-four brushes of one type and size only, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. The minimum bursting strength of the fiberboard shall be 200 p.s.i. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Brushes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The brushes covered by this specification are intended to be used for the application of glue.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and size of brush required (1.2).

(*c*) When preproduction sample is required (3.2).

(*d*) Identification marking required if other than specified (3.7).

(*e*) Applicable levels of packaging and packing (5.1 and 5.2).

6.3 **Supersession.** The following brushes contained in Emergency Federal Specification H-B-0292, dated August 22, 1951 were deleted from Federal Specification H-B-292a:

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Class 2.—Bristle and hair

Type F.—Flat

Size.—2 inch

Size.—3 inch

Type R.—Round

Size.—1 inch

Size.—1-3/16 inch

6.4 Illustration. The illustration shown herein as figure 1 is for the purpose of identification and is not intended to preclude any brush which otherwise meets the requirements of this specification.

6.5 Transportation description. Transportation description applicable to this item is as follows:

Rail:

Brushes, not otherwise indexed by name.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Brushes, not otherwise indexed.

Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—GL

Navy—S

Air Force—MOA

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